COSC300 GROUP 17 PROPOSAL

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School digital platform Solution

(Website + Mobile Application Integrated System)

The School Digital Platform is an integrated web and mobile solution designed to enhance communication, streamline academic management, and improve access to information for students, parents, teachers, and school administrators.

The platform consists of a responsive website and a mobile application that work together to provide real-time access to academic resources, school updates, and student performance information.

Key Objectives

Improve communication between school and families

Provide centralized access to homework, newsletters, and announcements

Showcase school achievements and activities

Digitize administrative and academic processes

Enhance student engagement through accessible digital tool

Problem Statement

Many schools rely on manual or disconnected systems to manage academic information and communication. As a result, parents and students may not receive timely updates about homework, announcements, or academic progress. In addition, teachers often struggle to identify students who are at risk of failing because important data—such as marks, attendance, and homework completion—is not analysed together. Without a system that can automatically analyse this data, struggling students may only be identified after poor academic performance has already occurred.

Solution Statement

This project proposes the development of a **School Digital Platform** that integrates a responsive website, a mobile application, and a centralized backend database. The platform will centralize school communication and academic data, allowing students, parents, teachers, and administrators to access relevant information in one place. The system will also incorporate **Artificial Intelligence** to analyse student academic data and **predict whether students are at risk of failing**, enabling teachers to intervene early and provide appropriate academic support.

1. Project Overview

The School Digital Platform will consist of:

A responsive website

A mobile application (Android & iOS)

A shared backend system (single database & admin panel)

The solution will provide a centralized communication, academic resource, and achievement management platform for students, parents, teachers, and administrators.

2. Solution Architecture Overview

2.1 Components

Public Website

About Us

Top Achievements

Weekly Announcements

Contact Information

Mobile Application

Same core features

Push notifications

Mobile-optimized user experience

Secure Portal System

Student login

Role-based access (Student, Teacher, Admin, Parent)

Homework & newsletter access

Admin Management System (CMS)

Central dashboard

Content publishing

User management

3. Technical Architecture

3.1 Backend

Centralized cloud-based server

Single database for website and app

REST API for mobile integration

Secure authentication system

Data encryption (SSL)

3.2 Frontend

Responsive web design (desktop, tablet, mobile browser)

Cross-platform mobile app (Android & iOS)

3.3 Hosting & Security

SSL certificate

Regular backups

POPIA compliance (for South Africa)

Secure file storage

3.4 AI Architecture

- Data collection module
- Data preprocessing
- Model training
- Model evaluation

- API integration **Data Collection Module**

The system will collect academic and behavioural data related to student performance. This includes:

- Student academic marks
- Attendance records
- Homework completion rates • Test and exam results

This data will be stored securely in the centralized database and used to train the AI model.

Data Preprocessing

Before the data is used for training the AI model, it must be cleaned and prepared. This step includes:

- Removing incomplete or inconsistent records
- Handling missing values
- Converting data into numerical format
- Normalizing academic performance data

This ensures that the machine learning model receives high-quality data for accurate predictions.

Model Training

The system will train a **Machine Learning model using Logistic Regression**.
The model will learn patterns from historical student data to identify indicators of academic risk.

The training process allows the system to understand relationships between:

- Marks
- Attendance
- Homework completion and student academic outcomes.

Model Evaluation

The trained AI model will be evaluated to measure how accurately it predicts student performance.

Evaluation metrics will include:

- Accuracy
- Precision
- Recall
- Confusion Matrix

These metrics help determine whether the model is reliable enough to be used in the system.

API Integration

Once the model is trained and evaluated, it will be integrated into the platform using an API.

This allows:

- The website and mobile app to send student data to the AI model
- The AI model to return predictions instantly
- The system to generate automatic risk classifications for students

4.Implementation phases

The implementation Phases of this project will follow the Full SDLC , using tools that are currently highly sought after and used in the industry, this will allow learners to have real project experience and know what to expect once they join teams in the corporate space

Implementation Phases

Phase 1: Requirements & Design

Phase 2: Backend Development

Phase 3: Website Development

Phase 4: Mobile App Development

Phase 5: Testing & Security Review

Phase 6: Deployment & Training

5. Artificial Intelligence Component

The School Digital Platform will incorporate a Machine Learning–based Student Performance Prediction Module.

This AI component will analyse:

Historical academic marks

Attendance records

Homework completion rates

Using a supervised learning algorithm (Logistic Regression), the system will classify students into:

Low Risk

Medium Risk

High Risk

The system will analyse student academic data and **predict whether a student is likely to pass or fail a subject before final examinations.**

Students classified as High Risk will be flagged by the system.

Teachers and administrators will be able to view these predictions through the platform dashboard, allowing them to identify struggling students early and provide appropriate academic support.

This predictive capability enables early intervention, helping schools improve student success rates and academic outcomes.

5.1 AI Tutor Expert

The platform will also include an AI Tutor Expert, which acts as an intelligent virtual learning assistant for students.

The AI tutor will help students understand difficult topics by providing:

- Simplified explanations
- Step-by-step guidance
- Examples related to the topic
- Learning suggestions and study tips

Students will be able to ask questions or upload learning content through the platform, and the AI tutor will respond with clear explanations, examples, and step-by-step guidance. This feature will help students learn independently and improve their understanding of complex subjects.

The AI Tutor Expert will also analyse student interaction patterns and suggest relevant study resources or topics that require additional practice. By providing personalized academic support, the AI tutor enhances the learning experience and encourages student engagement.

5.2 Teacher Notification and Alert System

The platform will provide an automated notification system to inform teachers about students identified as academically at risk.

When the AI model classifies a student as **High Risk**, the system will automatically generate an alert.

Teachers will receive this information through:

- **Teacher Dashboard Alerts** – High-risk students will be highlighted on the teacher's dashboard within the platform.
- **Push Notifications (Mobile App)** – Teachers using the mobile application will receive real-time notifications when a student is flagged.
- **Performance Reports** – Teachers will be able to view detailed student performance reports generated by the system.

These notifications will allow teachers to quickly identify struggling students and provide early academic intervention.

5.3 Additional AI Features

Artificial Intelligence will also be integrated into the platform to improve communication, academic monitoring, and data analysis. The AI components will analyse school data and generate useful insights for teachers and administrators.

1. **Student Performance Prediction:**
   A machine learning model will analyse marks, attendance records, homework completion, and test results to identify students at academic risk. The model will predict whether a student is likely to pass or fail before final examinations.
2. **Early Risk Classification:**
   Students will be categorized as **Low Risk, Medium Risk, or High Risk**, allowing teachers to intervene early and provide additional academic support.
3. **AI Content Summarization:**
   The system will automatically summarize long announcements, newsletters, and reports so that parents and students can quickly understand important information.
4. **AI Homework Assistant:**

Students will be able to receive explanations and guidance on homework tasks using AI-based assistance that helps them better understand difficult concepts.

5. **Attendance Pattern Analysis:**
   Artificial Intelligence will analyse attendance records and detect patterns that may indicate academic risk. Teachers will be alerted when irregular attendance may negatively affect a student's academic performance.

5.4 Key Data and Security Features

To ensure responsible and secure use of Artificial Intelligence within the platform, several data protection and security measures will be implemented.

Data Privacy

Student academic data such as marks, attendance records, and homework completion will be securely stored in the centralized database. This data will only be used for training and operating the prediction model within the platform and will not be shared with external systems.

Content Safety

The AI system will include safeguards to prevent the generation or display of harmful, inappropriate, or misleading content. All AI-generated summaries, reports, or recommendations will be monitored to ensure they remain relevant to the educational environment.

Human-in-the-Loop Review

AI-generated predictions and insights will serve as decision-support tools for teachers rather than automated decisions. Teachers and administrators will review the information before taking any academic actions or interventions.

Secure Data Storage

All student information will be protected through secure authentication systems, encrypted data storage, and role-based access control to ensure that only authorized users can access sensitive information.

6. Project Deliverables

The project will produce the following deliverables:

- Software Requirements Specification (SRS)
- System Design Documentation (UML Diagrams)
- Functional Website
- Functional Mobile Application
- AI Prediction Module
- Testing & Security Report
- Deployment Documentation
- Final Project Report & Presentation

7.Role of the Client/Sponsor

The client/sponsor will:

- Provide initial system requirements
- Validate system functionality
- Provide feedback during key development phases
- Participate in final system evaluation

No financial commitment is required from the sponsor.

A Detailed plan and ways of working document will be shared with the team and supervisor once a decision has been made.

Feel free to call me on 076 546 4236 for any questions you might have on this project.



Maemo Maponya

Director